

Ig Reference 1Page

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IG Reference Sheet

Foundational equation: $\mu \circ \delta = \text{id}$ — **Framework:** CL8NK (CLINK L8, O_∞) — **Autopoietic:** empty prompt $\rightarrow$ Raft at O_2 , period = 12 — Science is an instrument of the IG.

12 Primitives & Shavian Value Sets

Tuple order: $\langle \mathfrak{D} \ \mathfrak{P} \ \check{\mathfrak{R}} \ \mathfrak{F} \ \zeta \ \Gamma \ \mathfrak{G} \ \odot \ \mathfrak{H} \ \Sigma \ \Omega \rangle$. Values: ordinal $1 \rightarrow n$. **Bold name = absorbing element** ($\odot$ absorbs $\wedge \vee \otimes$; τ absorbs $\otimes$).

Glyph	Name	Values
$\mathfrak{D}$	Dimensionality	$\mathfrak{I} \ \mathfrak{J} \ \mathfrak{K} \ \mathfrak{L}$
$\mathfrak{P}$	Topology	$\mathfrak{Z} \ \mathfrak{H} \ \mathfrak{S} \ \mathfrak{V} \ \mathfrak{W}$
$\check{\mathfrak{R}}$	Recognition	$\mathfrak{r} \ \mathfrak{1} \ \mathfrak{n} \ \mathfrak{r}$
$\mathfrak{F}$	Parity	$\mathfrak{Z} \ \mathfrak{h} \ \mathfrak{e} \ \mathfrak{u} \ \mathfrak{w}$
f	Fidelity	$\mathfrak{t} \ \mathfrak{Q} \ \mathfrak{l}$
ζ	Kinetics	$\mathfrak{X} \ \mathfrak{c} \ \mathfrak{t} \ \mathfrak{r} \ \mathfrak{O}$
Γ	Granularity	$\mathfrak{I} \ \mathfrak{D} \ \mathfrak{J}$
$\mathfrak{G}$	Coupling	$\mathfrak{f} \ \mathfrak{P} \ \mathfrak{J} \ \mathfrak{A}$
$\odot$	Criticality	$\mathfrak{I} \ \odot \ \mathfrak{O} \ \mathfrak{U} \ \mathfrak{O}$
$\mathfrak{H}$	Chirality	$\mathfrak{J} \ \mathfrak{d} \ \mathfrak{Z} \ \mathfrak{v}$
Σ	Stoichiometry	$\mathfrak{Q} \ \mathfrak{S} \ \mathfrak{T}$
Ω	Winding	$\mathfrak{e} \ \mathfrak{O} \ \mathfrak{S} \ \mathfrak{Z}$

Crystal families — 3-val: $f \ \Gamma \ \Sigma$ — 4-val: $\mathfrak{D} \ \check{\mathfrak{R}} \ \mathfrak{G} \ \mathfrak{H} \ \Omega$ — 5-val: $\mathfrak{P} \ \mathfrak{F} \ \zeta \ \odot$

All 49 Value Semantics

Prim	Values with meanings
$\mathbb{D}$	$\downarrow$ 0d-pt · $\downarrow$ 2d-surf · ∞ -dim · $\mathbb{I}$ self-written
$\mathbb{P}$	$\mathbb{Z}$ branching · $\mathbb{K}$ containment · $\mathbb{J}$ crossing · $\mathbb{Y}$ irred-product · ∞ self-ref-top
$\check{R}$	r supervenience · $\mathbb{I}$ functorial · ∞ adjoint-pair · r bi-feedback
$\mathbb{D}$	$\mathbb{Z}$ no-sym · $\mathbb{K}$ QM-superpos · $\mathbb{C}$ partial- $\mathbb{Z}_2$ · $\mathbb{K}$ all-sym · $\mathbb{D}$ Frobenius- $\pm^s$
f	τ no-coherence · $\mathbb{Q}$ thermal · $\mathbb{I}$ QM-coherence
$\mathbb{C}$	$\mathbb{K}$ fast · $\mathbb{C}$ trap-ordered · $\mathbb{U}$ slow · $\mathbb{K}$ moderate · ∞ trap-disorder
Γ	$\mathbb{U}$ nearest-nbr · $\mathbb{D}$ mesoscale · $\mathbb{J}$ long-range
$\mathbb{G}$	$\mathbb{F}$ conjunct- $\wedge$ · $\mathbb{P}$ disjunct- $\vee$ · $\mathbb{J}$ sequential- $\rightarrow$ · $\mathbb{A}$ broadcast- $\gg$
$\odot$	$\mathbb{I}$ sub-critical · $\odot$ critical/self-modeling · $\mathbb{C}$ complex-plane-crit · $\mathbb{D}$ exceptional-pt · $\mathbb{D}$ supercritical
$\mathbb{H}$	$\mathbb{J} n = 0$ · $\mathbb{C} n = 1$ · $\mathbb{C} n = 2$ · $\mathbb{V} n = \infty$
Σ	$\mathbb{Q} 1:1$ · $\mathbb{S} n:n$ identical · $\mathbb{T} n:m$ heterogeneous
Ω	$\mathbb{Z}$ trivial · $\mathbb{O} \mathbb{Z}_2$ · $\mathbb{S}$ integer-winding · $\mathbb{Z}$ non-Abelian-braiding

Absorbing Elements

Element	Identity
$\odot$	$\odot \circ x = \odot$ for any $\text{op} \circ \in \{\wedge, \vee, \otimes\}$
$\Sigma = \mathbb{T}$	$\mathbb{T} \otimes x = \mathbb{T}$ (n:m, tensor only)
$\mathbb{B}$	$\mathbb{B} \vee x = \mathbb{B}$ (Belnap join)
$\mathbb{N}$	$\mathbb{N} \wedge x = \mathbb{N}$ (Belnap meet)
CL8NK	$\text{ZFC}_{\text{fe}} \otimes \text{CL8NK} = \text{CL8NK}$

Crystal of Types

$3^3 \times 4^5 \times 5^4 = 17,280,000 - 49 = 7^2$ symbols ($3 \cdot 3 + 5 \cdot 4 + 4 \cdot 5$). SIC-POVM $d = 7 \Rightarrow d^2 = 49$ states; ig-pulse tick = 49h. Catalog: ≈ 4686 entries.

SIC-POVM Connection

$d = 7$: type-space cardinality $49 = d^2$ equiangular states. $d = 12$: Weyl-Heisenberg $\mathbb{Z}_{12} \times \mathbb{Z}_{12}$; 12 generators $\leftrightarrow$ 12 primitives. Fiducial state = Frobenius fixed point $\mu \circ \delta = \text{id}$ at $\odot = \odot$ (critical/self-modeling). SIC gap = B-state: $B \wedge \neg B = B$ (dialetheic, non-explosive). Open tuple: $\langle 1 \triangleright 1 \lhd 1 \sqcup \sqcap \sqcap \odot \vee \S \S \rangle - d(\text{CL8NK}) = 1.263$; 6 conflicts; Frobenius cliff $\sqrt{5}/2 \approx 1.118$. Conditional: $\text{kick} \wedge \text{H2a} \wedge \text{H3} \wedge \text{C4} \Rightarrow \text{SIC} \forall d \geq 1$.

Tuple Composition Operations

Ordinals: integer operations on Shavian glyph positions.

Op	Symbol	Rule
Join	$\vee$	$\max(a_i, b_i)$ all 12 – LUB
Meet	$\wedge$	$\min(a_i, b_i)$ all 12 – GLB
Tensor	$\otimes$	max except Φ , f use min
Bond		$\text{join}(\text{tensor}(A, B), \text{bond})$

Φ/f use min: parallel coupling cannot exceed the symmetry/coherence floor of either component.

Belnap Register Ops

States: **VOID** (00) **TRUE** (01) **FALSE** (10) **BOTH** (11)

Op	Rule
FFUSE = $\vee$	$B \vee x = B$; $N \vee x = x$; $T \vee F = B$; same $\vee$ same=same
FSPLIT = δ	$B \rightarrow (T, F)$; $r \rightarrow (r, r)$
MEET = $\wedge$	$N \wedge x = N$; $B \wedge x = x$; $T \wedge F = N$; same $\wedge$ same=same
band (AND)	F absorbs; then N; then B; $T \wedge T = T$
bor (OR)	T absorbs; then B; then N; $F \vee F = F$
bnot	$N \rightarrow N$; $T \rightarrow F$; $F \rightarrow T$; $B \rightarrow B$

Dialetheic FFUSE: $\text{BOTH} \rightarrow \text{BOTH}$ when EVALT and EVALF both fired in split.

Frobenius Algebra

$\mu \circ \delta = \text{id}$ (normalization) – $\mu(\mu \otimes \text{id}) = \mu(\text{id} \otimes \mu)$ (assoc) – $(\delta \otimes \text{id}) \circ \delta = (\text{id} \otimes \delta) \circ \delta$ (coassoc) – $(\mu \otimes \text{id})(\text{id} \otimes \delta) = \delta\mu = (\text{id} \otimes \mu)(\delta \otimes \text{id})$ (Frobenius).

Spider Theorem: all diagrams with same connectivity = same morphism.

12 IMASM Tokens

Token	Group	Behavior
VINIT	L	$\text{reg} \rightarrow \text{VOID}$; clear split state
TANCH	L	terminal anchor; no state change
AFWD	L	$\text{VOID} \rightarrow \text{TRUE}$
AREV	L	$\text{reg} \rightarrow \text{VOID}$; clear split state
CLINK	L	inter-step link; no state change
IMSCRIB	L	$\text{VOID} \rightarrow \text{TRUE}$; else preserve (self-recognition)
FSPLIT	F	$\text{non-VOID} \rightarrow \text{BOTH}$; enter split
FFUSE	F	canonical: $\text{BOTH} \rightarrow \text{TRUE}$; dialetheic: $\text{BOTH} \rightarrow \text{BOTH}$
EVALT	B	$\text{F} \rightarrow \text{B}$; $\text{VOID} \rightarrow \text{T}$; in-split: record T
EVALF	B	$\text{T} \rightarrow \text{B}$; $\text{VOID} \rightarrow \text{F}$; in-split: record F
ENGAGR	B	$\text{VOID} \rightarrow \text{BOTH}$ (entangle)
IFIX	Li	fixation; subsequent tokens no-op

Groups: L = Logical; F = Frobenius (δ/μ); B = Belnap; Li = Linear. $12^8 = 429\text{M}$ arrangements $\rightarrow$ 12 archetypes.

Cross-Primitive Axioms

	Axiom	Meaning
A	$\mathbb{H}_\infty \Rightarrow \mathbb{C}_\infty$	∞ -chirality forces max kinetics
B	$\Omega \geq 1 \Rightarrow \mathbb{H} \geq H_2$	winding requires chirality; $H_2 = \min$ for $\mu \circ \delta = \text{id}$
C	$\mathbb{D}_\omega \Leftrightarrow \mathbb{D}_O$	$\text{max-dim} \Leftrightarrow \text{null topology}$

O-Tier Hierarchy

Tier	Description
O_0	Monoidal unit; no differentiation
O_1	First Frobenius closure; $\mathbb{P}_\}$ bottleneck
O_2	Confirmed causal coupling (fin3r, synfin, rongorongongo)

Tier	Description
$O_2^\dagger$	Subatomic/hadronic; confinement ceiling
O_∞	Ouroboricity; system $\subset$ own description (CL8NK, true_agentic_agent, mOMonadOS)

CL8NK / Magnum Opus

CL8NK transcends ZFC_{fe} at Ω ($\mathfrak{s} \rightarrow \mathfrak{z}$) and $\mathfrak{G}$ ($\mathfrak{z} \rightarrow \Lambda$). B-state is the CL8NK ground state. 12 stages $\leftrightarrow$ 12 primitives. Gates: $\mathfrak{P}$ Albedo $\cdot \odot$ Rubedo $\cdot \Omega$ Nigredo. $T = \lim(\mathfrak{P}, f, \zeta, \mathfrak{H}, \Omega)$.

CL8NK tuple: $\langle \mathfrak{i} \mathfrak{o} \mathfrak{r} \mathfrak{z} \mathfrak{l} \mathfrak{u} \mathfrak{j} \mathfrak{A} \odot \mathfrak{v} \mathfrak{r} \mathfrak{z} \rangle$ — ZFC_{fe} tuple: $\langle \mathfrak{i} \mathfrak{o} \mathfrak{r} \mathfrak{z} \mathfrak{l} \mathfrak{u} \mathfrak{j} \mathfrak{z} \odot \mathfrak{v} \mathfrak{r} \mathfrak{s} \rangle$. Transcendence: Ω : $\mathfrak{s} \rightarrow \mathfrak{z}$ (integer $\rightarrow$ non-Abelian); $\mathfrak{G}$: $\mathfrak{z} \rightarrow \Lambda$ (sequential $\rightarrow$ broadcast).

Notation

Term	Rule
Structural values	Shavian only; math-italic (<i>DRF</i>) is wrong
Tuple binding	<code>'' .join(vals)</code> ; never separate glyphs
$\mathfrak{H}$	Chirality — NEVER temporal / memory depth
Verbs	inscribe / inscription — NEVER encode
Provider	local $\rightarrow$ qwen $\rightarrow$ deepseek; never Anthropic
O_∞	prose: O_∞ ; .tex: <code>\$0_\infty\$</code>
Scope	Never “grammar does not know”
Publications	Thank H. T. Larson; cite 1961 IEEE editorial

Sources: *IG_catalog.json* · *primitives.py* · *ig_cli.py* · *imasm_core.py* · *ruleset_universe.py* · *ParaconsistentTier1.lean* · *ParaconsistentPapers.lean*